

TIPMAP: SV-centered TE-SV discovery from pangenome variation

Pangenome SV evidence

VCF-aware parsing of PanPop and query evidence

PanPop merged SV VCF multi-sample SV genotypes

CHROM	POS	REF	ALT	FORMAT	CM104	JM47
Chr1A	1157425	TCT...	A,ATC...	GT	1/1	2/2
Chr1A	722313	GAA...	G	GT	0/0	1/1

INS: ALT **DEL: REF** (skip 0/0 samples) multi-allelic aware

Pairwise-to-reference VCFs query-specific true ALT

CHROM	POS	REF	ALT	INFO	JM47
Chr1A	1444154	A	AAAT...	ASM_Start=...	1

—  — ALT in query
ASM_* = query coords

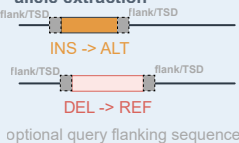
Reference and query FASTA

>CS Chr1A >JM47 Chr1A
ATGCTAAC... ...TSD|TE|TSD

1 Extract SV alleles

INS=ALT; DEL=REF; skip 0/0

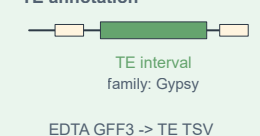
allele extraction



2 Annotate TE evidence

rmDup + EDTA; or existing GFF3

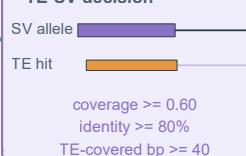
TE annotation



3 Classify TE-SVs

BLASTN coverage + identity

TE-SV decision



Final TIP map

TE alleles + rewritten GT

TIP-map VCF

ID	ALT kept	rewritten GT
SV001	ALT1,ALT3	1/1 0/0 2/2

non-TE ALT genotypes -> 0/0

TIP presence matrix



Allele support

TE family
coverage + identity

TE-support table

SV_ID	Chr1A_...
allele	ALT1
family	Gypsy
coverage	0.71
identity	86%

Allele-aware genotype rewriting

ALT1=TE; ALT2=non-TE; ALT3=TE

1/1 -> 1/1 2/2 -> 0/0 3/3 -> 2/2

TE allele rule

coverage >= 0.60; identity >= 80%; TE-covered bp >= 40